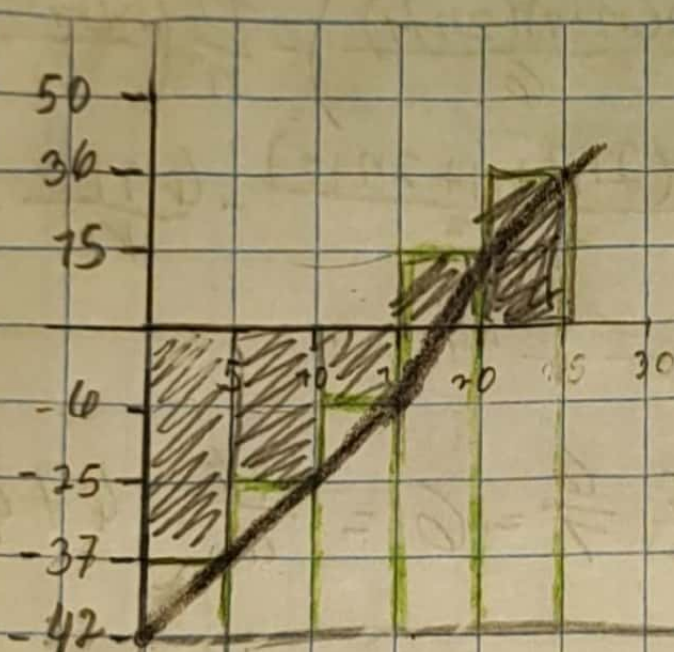


Tarea Sección 5.2

7)



$$5 * f(0) = 5 * -42 = -210$$

$$5 * f(5) = 5 * -37 = -185$$

$$5 * f(10) = 5 * -25 = -125$$

$$5 * f(15) = 5 * -10 = -50$$

$$5 * f(20) = 5 * 15 = 75$$

$$\underline{-415}$$

$$5 * f(5) = -185$$

$$5 * f(10) = -125$$

$$5 * f(15) = -50$$

$$5 * f(20) = 75$$

$$5 * f(25) = 180$$

$$\underline{-85}$$

$$21) \int_{-1}^5 (1+3x) dx$$

$$\Delta x = \frac{5 - (-1)}{n} = \frac{6}{n} \quad x_i = -1 + \frac{6i}{n} = \frac{6i}{n} - 1$$

$$f(x_i) = 1 + 3\left(\frac{6i}{n} - 1\right) = 1 + 3\left(\frac{6i}{n} - 1\right) = 1 + \frac{18i}{n} - 3 = \frac{18i}{n} - 2$$

$$\int_{-1}^5 (1+3x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{18i}{n} - 2\right) \left(\frac{6}{n}\right)$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{108i}{n^2} - \frac{12}{n}\right)$$

$$= \lim_{n \rightarrow \infty} \frac{108}{n^2} \sum_{i=1}^n i - \frac{12}{n} \sum_{i=1}^n 1$$

$$= \lim_{n \rightarrow \infty} \frac{108}{n^2} \left(\frac{n(n+1)}{2}\right) - \frac{12}{n} (n)$$

$$= \lim_{n \rightarrow \infty} \frac{54n + 54}{n} - 12 = \frac{54n + 54 - 12n}{n} = \frac{42n + 54}{n}$$

$$= \lim_{n \rightarrow \infty} \frac{42}{1} = 42$$

$$23) \int_{-2}^0 (x^2 + x) dx$$

$$\Delta x = \frac{0 - (-2)}{n} = \frac{2}{n} \quad x_i = -2 + \frac{2i}{n}$$

$$f(x_i) = \left(\frac{2i}{n} - 2\right)^2 + \left(\frac{2i}{n} - 2\right) = \left(\frac{4i^2}{n^2} - \frac{8i}{n} + 4\right) + \left(\frac{2i}{n} - 2\right) = \frac{4i^2}{n^2} - \frac{6i}{n} + 2$$

$$\int_{-2}^0 (x^2 + x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{4i^2}{n^2} - \frac{6i}{n} + 2\right) \left(\frac{2}{n}\right) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{8i^2}{n^3} - \frac{12i}{n^2} + \frac{4}{n}$$

$$\lim_{n \rightarrow \infty} \frac{8}{n^3} \sum_{i=1}^n i^2 - \frac{12}{n^2} \sum_{i=1}^n i + \frac{4}{n} \sum_{i=1}^n 1 = \frac{8}{n^3} \left(\frac{n(n+1)(2n+1)}{6} \right) - \frac{12}{n^2} \left(\frac{n(n+1)}{2} \right) + \frac{4}{n} (n)$$

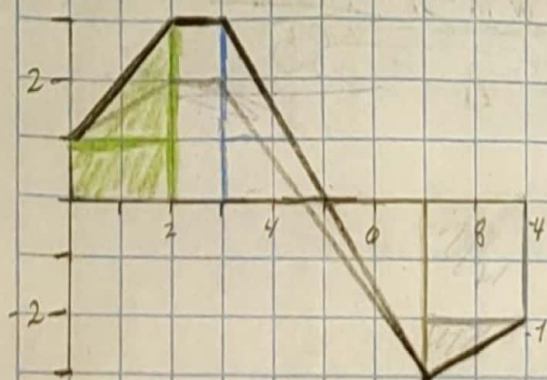
$$\frac{4(n+1)(2n+1)}{3n^2} - \frac{6(n+1)}{n} + 4 = \frac{4(2n^2+n+2n+1)}{3n^2} - \frac{6n+6}{n} + 4 =$$

$$\frac{8n^2+12n+4}{3n^2} - \frac{6n+6}{n} + 4 =$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{7(6n+12)}{6n} = \frac{76}{6} - \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{6}{7} = 6 = \frac{76}{6} - 6 + 4 = \frac{2}{3}$$

33) a) $\int_0^2 f(x) dx$ c) $\int_5^7 f(x) dx$ 10 - 3 = 7

b) $\int_0^5 f(x) dx$ d) $\int_0^9 f(x) dx$



a) $\int_0^2 \left(\frac{1}{2}\right)(bh) = \frac{1}{2}(2) = 1$

b) $\int_0^5 4 + 3 + 3 = 10$

c) $\int_5^7 \frac{1}{2} = \frac{1}{2} = -3$

d) $\int_0^9 10 - 3 - 5 = 2$

35) $\int_2^4 (1-x) dx$ $\Delta x = \frac{4-2}{n} = \frac{2}{n}$ $x_i = 2 + \frac{2i}{n}$ $a=2$ $b=4$

$$f(x_i) = 1 - \left(2 + \frac{2i}{n}\right) = -1 - \frac{2i}{n}$$

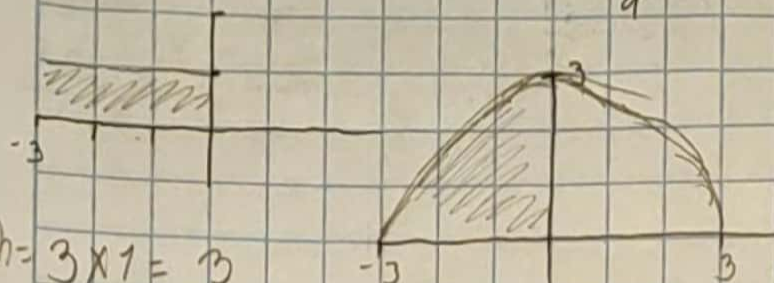
$$\int_2^4 (1-x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(-1 - \frac{2i}{n}\right) \left(\frac{2}{n}\right) = -\frac{2}{n} - \frac{4i}{n^2}$$

$$= \lim_{n \rightarrow \infty} -\frac{2}{n} \sum_{i=1}^n 1 - \frac{4}{n^2} \sum_{i=1}^n i = -\frac{2}{n}(n) - \frac{4}{n^2} \left(\frac{n(n+1)}{2}\right)$$

$$= -2 - \frac{2(n+1)}{n} = -2 - \left(\frac{2n+2}{n}\right) = -2 - \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2n+2}{n} =$$

$$= -2 - \frac{2}{1} = -2 - 2 = -4$$

$$37) \int_{-3}^0 (1 + \sqrt{9-x^2}) dx = \underbrace{\int_{-3}^0 1 dx}_a + \underbrace{\int_{-3}^0 \sqrt{9-x^2} dx}_b$$



$$\frac{\pi r^2}{4} = \frac{3^2 \pi}{4} = \frac{9\pi}{4}$$

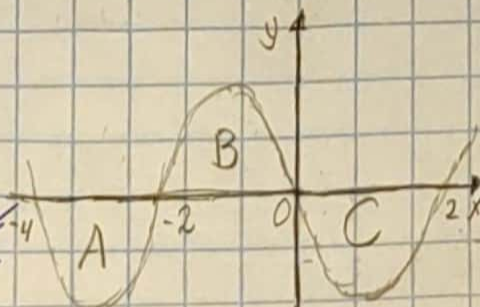
$$bh = 3 \times 1 = 3$$

$$\int_{-3}^0 (1 + \sqrt{9-x^2}) dx = 3 + \frac{9\pi}{4} = 70.07$$

$$53) \int_{-4}^2 [f(x) + 2x + 5] dx$$

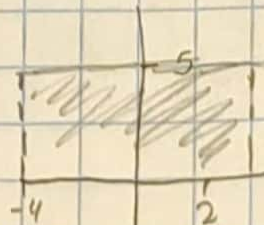
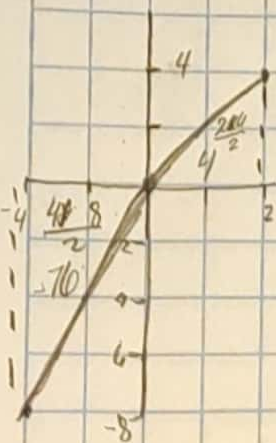
$$\int_{-4}^2 f(x) dx + \int_{-4}^2 2x dx + \int_{-4}^2 5 dx$$

$$-3 + [-16 + 4] + 30 = 15$$



$$A = B = C = 3$$

$$-3 + 3 - 3 = -3$$



$$6 \times 5 = 30$$